

Lab Sheet 6: HCI & Design Thinking

Objectives

By completing today's workshop, you will:

- Apply Human-Computer Interaction (HCI) principles to design a User Interface (UI).
 - Create a low-fidelity wireframe prototype of your Ground Control Station dashboard.
 - Conduct a Heuristic Evaluation of your prototype with human peers and an AI assistant.
 - Build an interactive "Limited functionality simulation" using frontend code or Generative AI tools.
 - Perform an automated Accessibility audit to ensure LEPSI/WCAG compliance.
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Introduction: Prototyping Your Dashboard

In your Assessment 1 project, you are tasked with building a web-based Ground Control Station to interact with a Virtual Robot. A common mistake in software engineering is jumping straight into writing HTML, CSS, and JavaScript before knowing what the user actually needs. Today, we will use the **Design Thinking** methodology to Prototype and Test your interface before you write any production code.

Task 1: Wireframing the Web Dashboard

Goal: Design the layout of your Ground Control Station and map out one specific feature in detail.

The Concept: Wireframes are low-fidelity visual representations of your interface. They focus on layout, information architecture, and user flow rather than colors and typography.

Action:

1. Grab a pen and paper, or open a digital wireframing tool like Figma or Excalidraw.
2. **The Main Dashboard:** Sketch the primary view of your Ground Control Station. Ensure you include:
 - A navigation bar/menu.
 - A display for the Robot's current status (Battery, Status, X/Y Position).
 - A visual indication of the current user's Role (Commander, Viewer, etc.).

3. **Deep-Dive Feature:** Pick one complex feature (e.g., The "Move Robot" control panel or the "Mission Audit Log") and draw a detailed, zoomed-in sequence of how the user interacts with it.
4. *Check:* Did you apply Fitts's Law? Are your primary action buttons (like "Move" or "Emergency Stop") large and easily accessible?

Task 2: Peer and AI Usability Evaluation

Goal: Evaluate the effectiveness of your design using established heuristics before building it.

The Concept: We will simulate a Heuristic Evaluation. You will gather feedback from two different sources: your human peers (who understand the project context) and a multimodal AI model.

Action:

1. **Peer Evaluation:** Show your sketches to at least **two other students** in the room. Ask them to evaluate your UI against Shneiderman's Golden Rules (e.g., "Is there informative feedback when I click 'Move'?", "Is there an easy reversal/undo option?"). Write down their feedback.
2. **AI Evaluation:** Take a clear photo or screenshot of your wireframe.
3. Open a multimodal AI (with your student account, you have free access to GitHub Copilot Pro: <https://github.com/copilot>) and upload your image.
4. Use the following prompt to ask the AI for a formal critique:

You are an expert UX/UI Researcher. I have uploaded a low-fidelity wireframe of a Ground Control Station dashboard used to pilot a Virtual Robot. Please conduct a Heuristic Evaluation of this interface based on Norman's 7 Principles and Shneiderman's 8 Golden Rules. Identify at least 3 usability issues and suggest specific improvements.

5. **Refine:** Compare the feedback from your peers and the AI. Redraw or update your wireframe to fix the identified usability issues.

Task 3: Limited Functionality Simulation (Interactive Prototype)

Goal: Create a working, interactive prototype of your dashboard in the browser.

The Concept: In HCI, a "Limited functionality simulation" is a prototype that looks and feels like the final product but doesn't have a real backend connected to it. You must convert your refined wireframe from Task 2 into actual frontend code (HTML/CSS/JS).

Action (Choose Path A or Path B):

→Path A: For students with Web Development Experience

- Manually code your dashboard based on your wireframe.

- Instead of writing all the CSS from scratch, we highly recommend using a **UI Framework**. Frameworks provide pre-built, standardized, and accessible components (buttons, navbars, grid systems).
- **Recommended Frameworks:** Bootstrap, Tailwind CSS, or your favourite one.

→**Path B: For beginners (AI-Assisted)**

- If you have zero frontend experience, you can use Generative AI tools (with your student account, you have free access to GitHub Copilot Pro: <https://github.com/copilot>) to translate your sketch into code.
- Upload your final wireframe and provide a prompt describing the styling you want, such as:

Act as an expert Frontend Developer. Look at the wireframe I uploaded for a Robot Ground Control Station. Please generate a single-file "Limited functionality simulation" (HTML, CSS, and vanilla JavaScript) of this dashboard using the Bootstrap 5 framework. Make the styling modern, using a dark-mode theme suitable for a robotics control room. Make the "Move" button interactive so that clicking it shows a mock success notification.

- Of course, the AI-generated code will not be perfect. You will need to review it, understand it, and make manual adjustments to ensure it matches your design and is functional. **Importantly, as with any AI usage, document your process as detailed in the assessment brief.**

Final Step for both paths: Save the resulting code into your project repository and open it in your web browser to test. This will serve as the foundational starting point for your Assessment 1 frontend!

Task 4: Accessibility Audit

Goal: Ensure your interactive prototype is compliant with LEPSI and WCAG accessibility standards without relying on specific browser extensions.

The Concept: A beautiful interface is useless if visually impaired or motor-impaired users cannot navigate it. We will conduct a three-part browser-agnostic accessibility audit focusing on Structure, Contrast, and Keyboard Navigation.

Action:

1. **Structural Audit (W3C):** Go to the W3C Markup Validation Service. Select "Validate by File Upload" and upload your html file.
 - Look for structural accessibility errors, such as missing `alt` attributes on images, missing `<label>` tags for inputs, or missing `<title>` tags.
2. **Color Contrast Audit (WebAIM):** Open the WebAIM Contrast Checker in your browser.
 - Enter the exact HEX color codes of your dark-mode background and your text.

- Does it pass the WCAG AA standard (a ratio of at least 4.5:1)? If it fails, adjust your CSS colors until it passes.
3. **Manual Keyboard Audit:** Open your HTML file in *any* web browser. Take your hand off your mouse/trackpad.
 - Try to navigate to your "Move Robot" button using **ONLY** the Tab key, and "click" it using the Enter key.
 - Is it visually obvious which element currently has keyboard focus? If not, you need to add `:focus` styles to your CSS!
 4. **Fix the Code:** Update your HTML/CSS to resolve any issues discovered during these three tests.

Further Resources

- **Bootstrap Documentation:** <https://getbootstrap.com/> - *The most popular UI framework for rapidly building responsive layouts.*
- **W3C Markup Validator:** <https://validator.w3.org/> - *The official tool to check your HTML for structural and accessibility errors.*
- **WebAIM Contrast Checker:** <https://webaim.org/resources/contrastchecker/> - *Ensure your UI colors are legally compliant with WCAG guidelines.*
- **10 Usability Heuristics (NN/g):** nngroup.com/articles/ten-usability-heuristics/